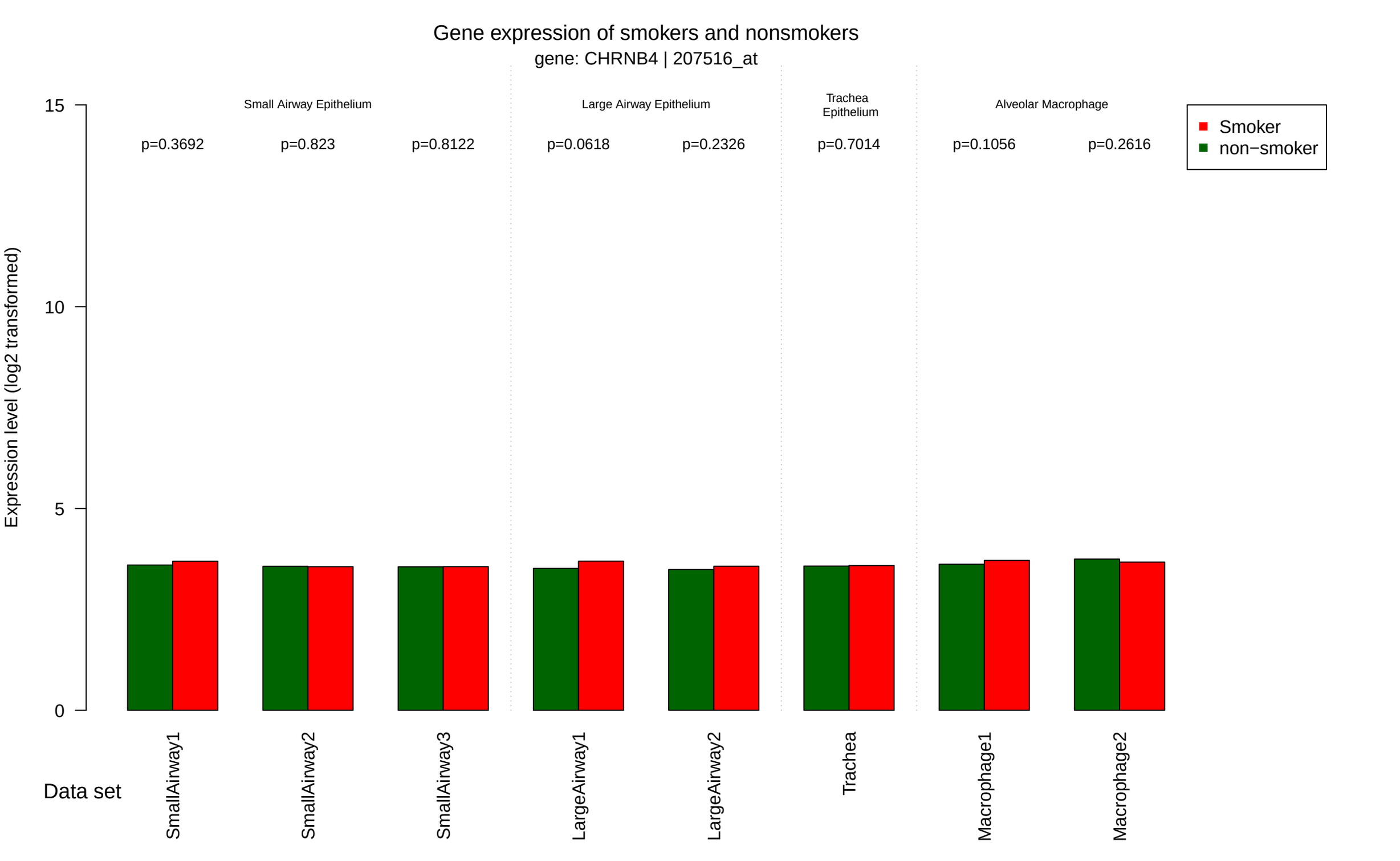
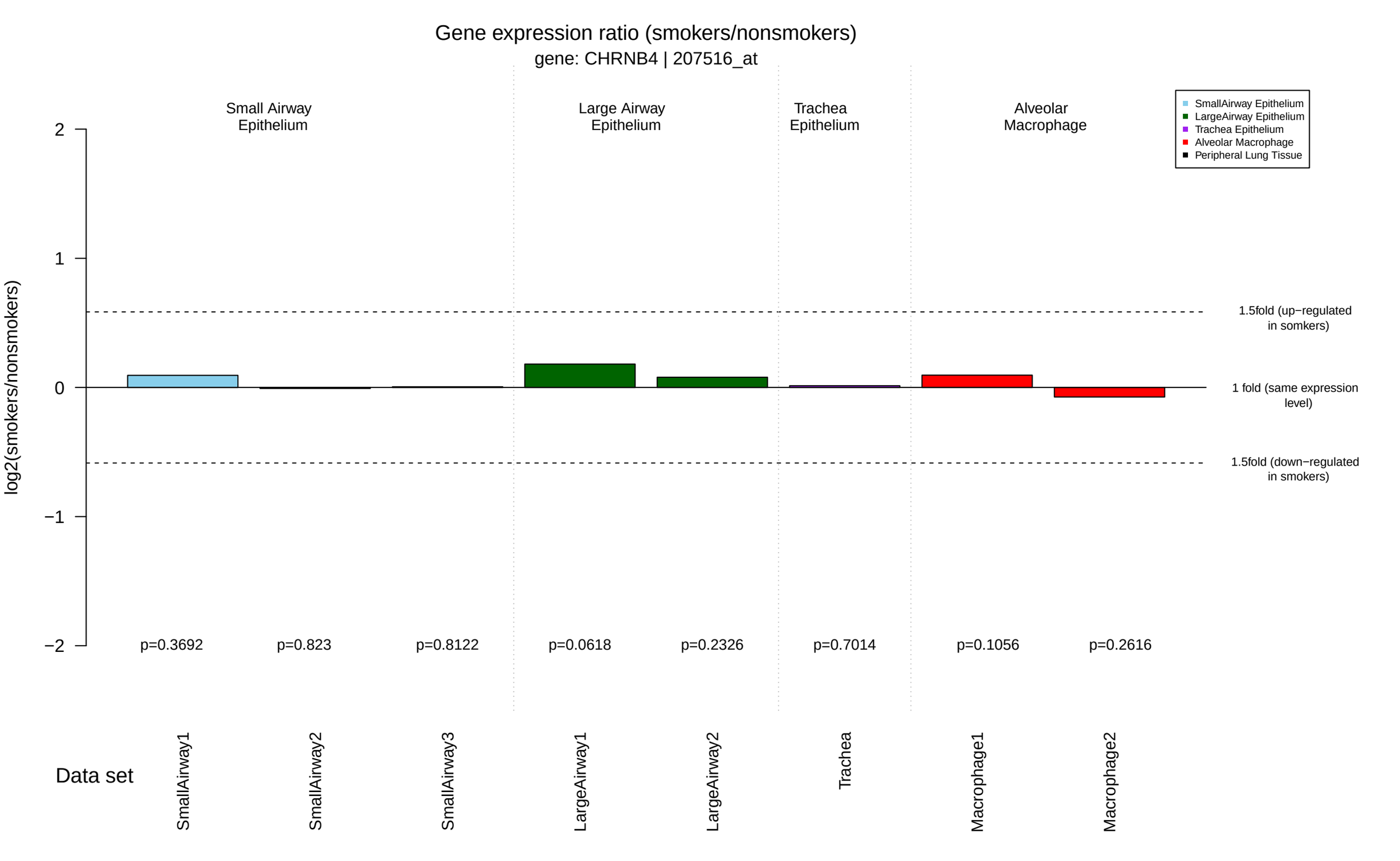
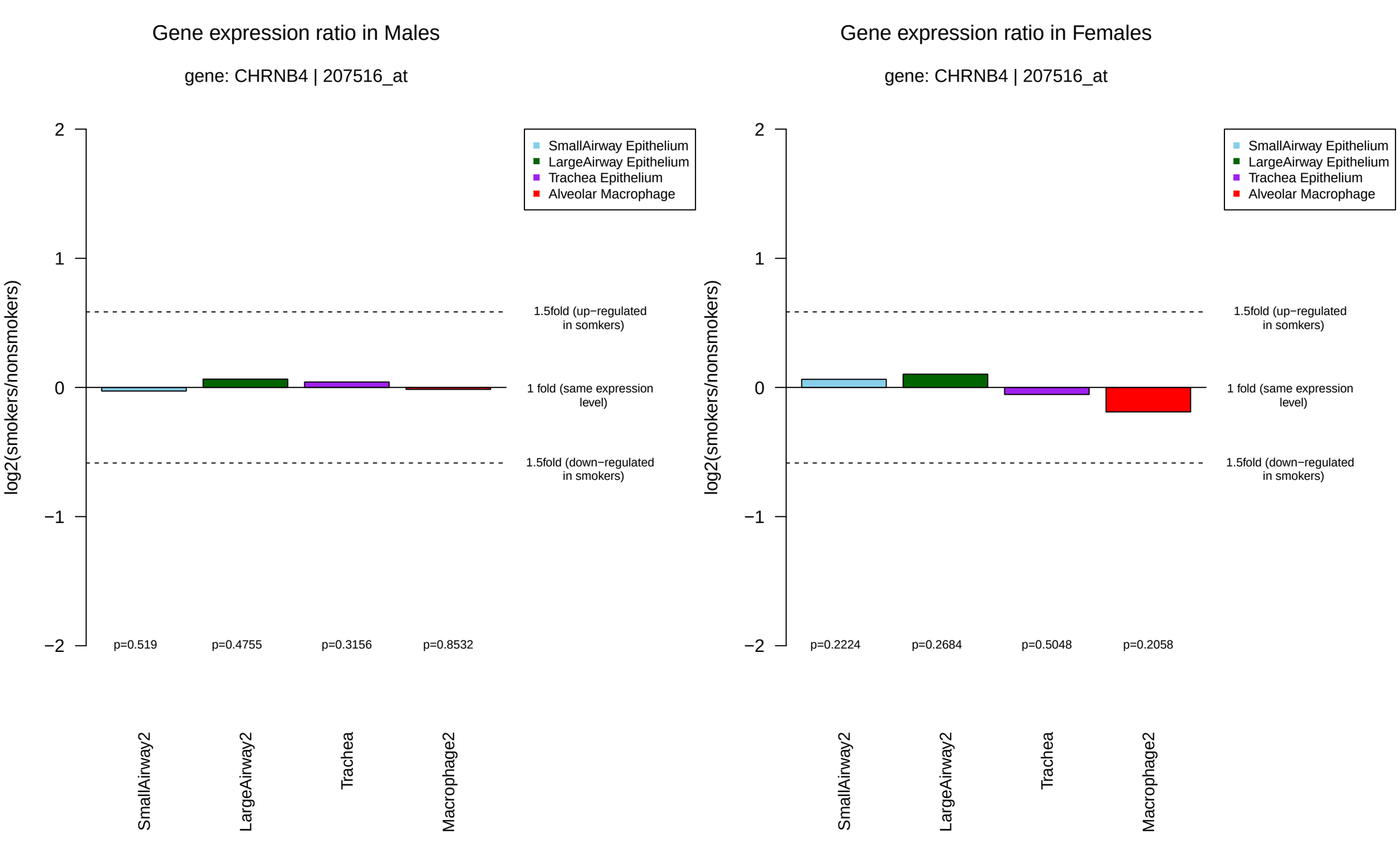
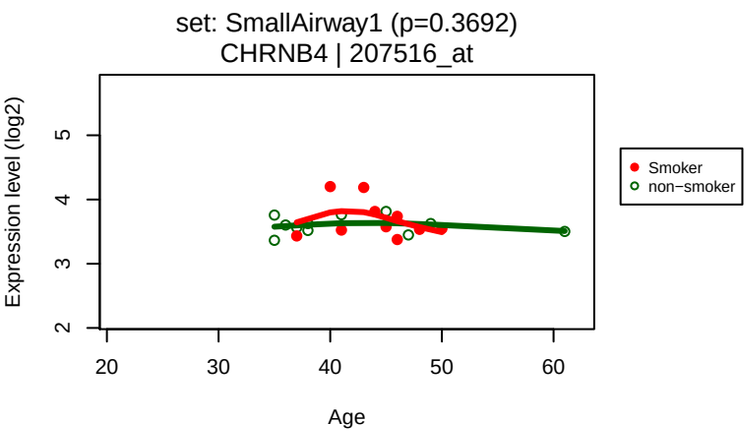
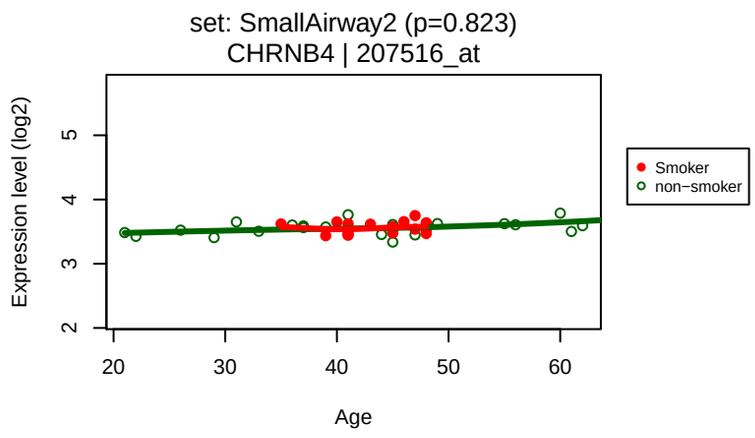
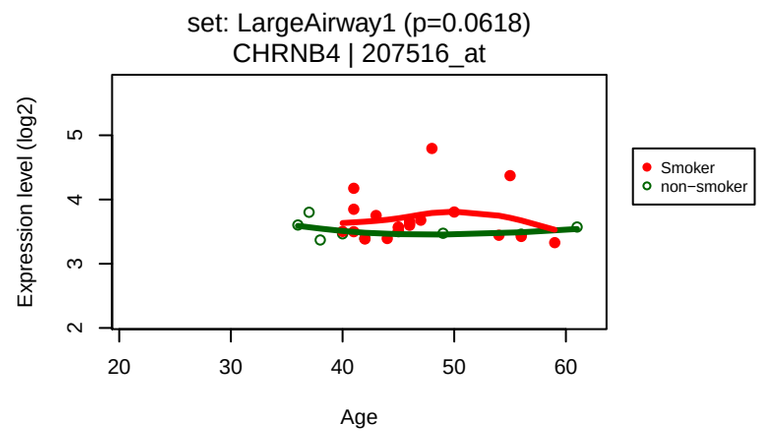

Gene expression of smokers and nonsmokers
gene: CHRNB4 | 207516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3692
p=0.823
p=0.8122
p=0.0618
p=0.2326
p=0.7014
p=0.1056
p=0.2616
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

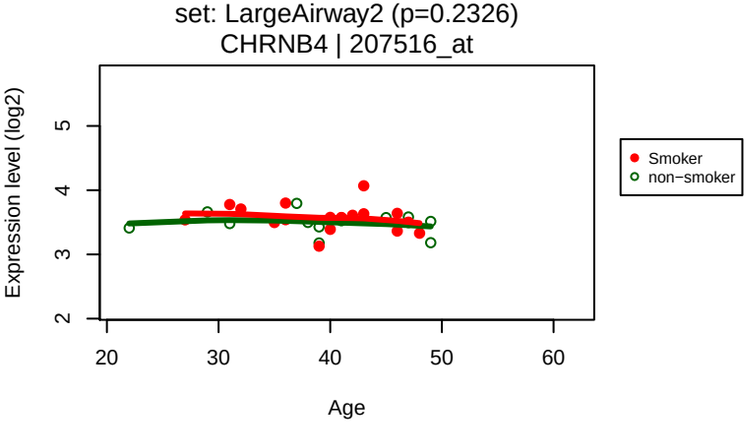

Gene expression ratio (smokers/nonsmokers)
gene: CHRNB4 | 207516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3692
p=0.823
p=0.8122
p=0.0618
p=0.2326
p=0.7014
p=0.1056
p=0.2616
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

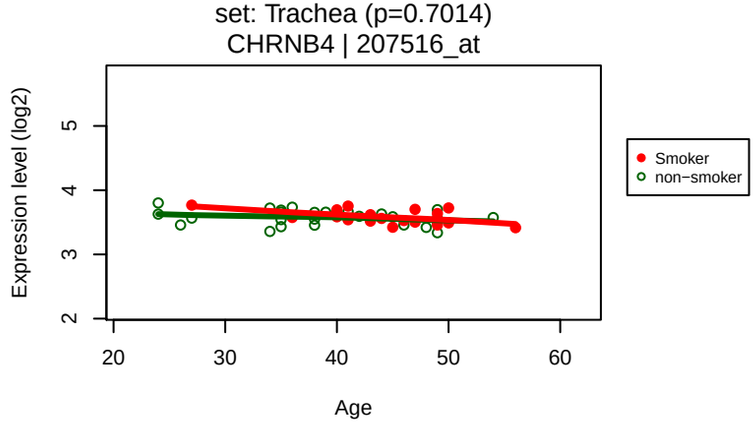

Gene expression ratio in Males
gene: CHRNB4 | 207516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.519
p=0.4755
p=0.3156
p=0.8532
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNB4 | 207516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2224
p=0.2684
p=0.5048
p=0.2058
SmallAirway2
LargeAirway2
Trachea
Macrophage2

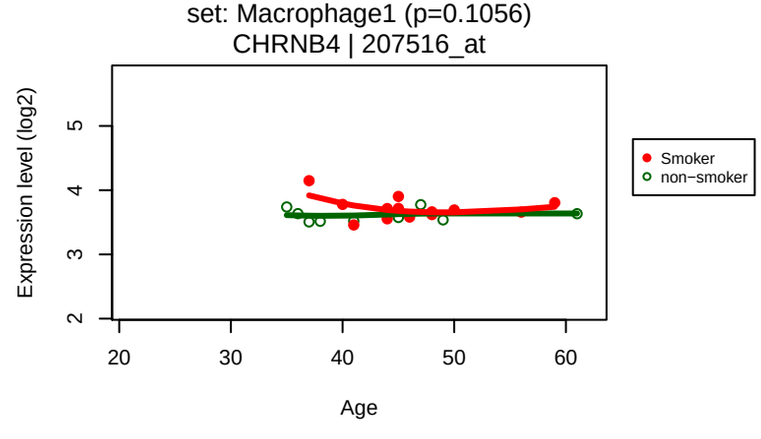

set: SmallAirway1 (p=0.3692)
CHRNB4 | 207516_at
Expression level (log2)
Age
Smoker
non−smoker

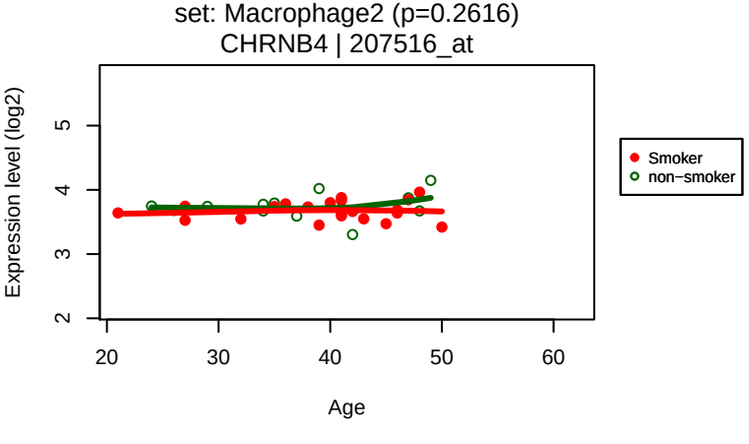

set: SmallAirway2 (p=0.823)
CHRNB4 | 207516_at
Expression level (log2)
Age
Smoker
non−smoker

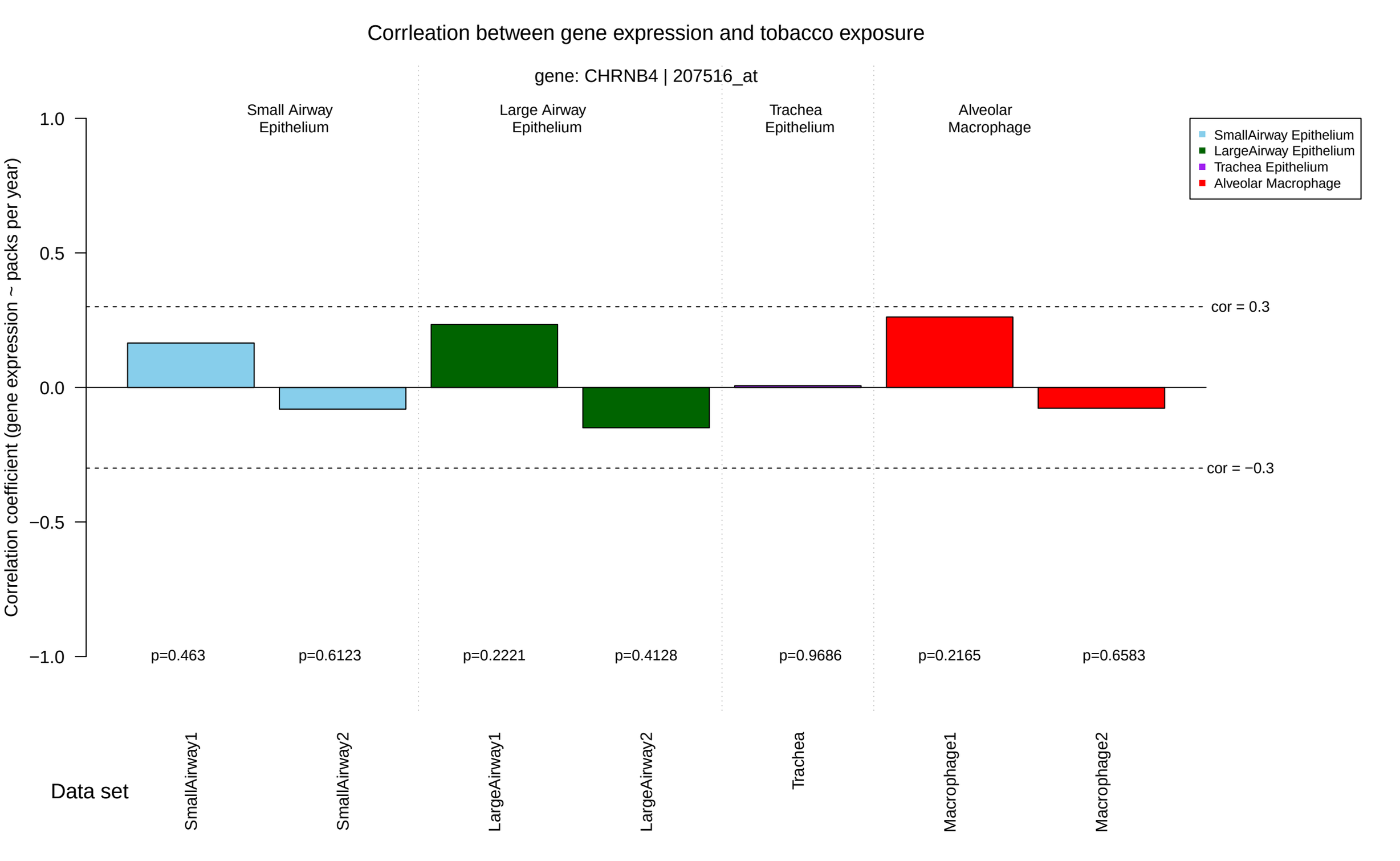

set: LargeAirway1 (p=0.0618)
CHRNB4 | 207516_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2326)
CHRNB4 | 207516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7014)
CHRNB4 | 207516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1056)
CHRNB4 | 207516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2616)
CHRNB4 | 207516_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNB4 | 207516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.463
p=0.6123
p=0.2221
p=0.4128
p=0.9686
p=0.2165
p=0.6583
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2